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Course: Advance Programming

Q1. Write a Java program using ArrayList to store book titles. Add at least 5 books and search for books whose title contains a given word

```
Users > kumar > .vscode > ap.java > ...
1  import java.util.ArrayList;
2  import java.util.Scanner;
3
4  class BookSearch {
5
6      public static void main(String[] args) {
7
8          ArrayList<String> books = new ArrayList<String>();
9
10         books.add("The Alchemist");
11         books.add("Harry Potter");
12         books.add("Think and Grow Rich");
13         books.add("The Power of Habit");
14         books.add("Atomic Habits");
15
16         Scanner sc = new Scanner(System.in);
17
18         System.out.print("Enter word to search: ");
19         String word = sc.nextLine().toLowerCase();
20
21         boolean found = false;
22
23         System.out.println("\nMatching books:");
24
25         for (int i = 0; i < books.size(); i++) {
26             String book = books.get(i);
27
28             if (book.toLowerCase().contains(word)) {
29                 System.out.println(book);
30                 found = true;
31             }
32         }
33
34         if (!found) {
35             System.out.println("No matching books found.");
36         }
37
38         sc.close();
39     }
40 }
41
```

Output:

```
PS C:\> & 'C:\Program Files\Java\jre1.8.0_461\bin\java.exe' '-cp' 'C:\Users\k
java-project\bin' 'BookSearch'
Enter word to search: A

Matching books:
The Alchemist
Harry Potter
Think and Grow Rich
The Power of Habit
Atomic Habits
PS C:\> ^C
PS C:\>
PS C:\> cd 'c:\\'; & 'C:\Program Files\Java\jre1.8.0_461\bin\java.exe' '-
\jdt_ws\jdt.ls-java-project\bin' 'BookSearch'
Enter word to search: b

Matching books:
The Power of Habit
Atomic Habits
PS C:\> █
```